

Group work set 2: Game Theory, Market Failure

Q1. Two firms, Arnold GmbH and Bobby GmbH, are both producing cars. They must each decide, simultaneously, between producing large cars and small cars. If both firms produce large cars, they each earn a profit of 5 Million EUR. If Arnold produces large cars and Bobby produces small cars, then Arnold earns 8 Million EUR, but Bobby earns only 2 Million EUR. If Bobby produces large cars and Arnold produces small cars, then Bobby earns 7 Million EUR and Arnold earns 3 Million EUR. If both produce small cars, they each earn 1 Million EUR.

(a) Construct a payoff table.

	Bobby: large	Bobby: small
Arnold: large	(5, 5)	(8, 2)
Arnold: small	(3, 7)	(1, 1)

(b) For each of the cells in the table, state whether it is (1) Pareto-efficient and (2) Kaldor-Hicks efficient.

All cells except for (small, small) are both Pareto efficient and KH-efficient: they yield the maximum payoff of 10 Million Dollar. The cell (small, small) is neither Pareto efficient nor KH-efficient.

(c) What is the Nash Equilibrium (or Nash Equilibria) of this game?

Arnold has a dominant strategy of producing large cars (as $5 > 3$ and $8 > 1$). Bobby has a dominant strategy to produce large cars as well (as $5 > 2$ and $7 > 1$). Hence, they both produce large cars – the equilibrium is (large, large).

(d) Suppose that Bobby moves first. Arnold observes Bobby's decision, and moves second. What is the Sub-game perfect Nash Equilibrium (or equilibria) of the game now?

As both parties have a dominant strategy, converting the game from static to dynamic does not affect the equilibrium. It is still (large, large).

(e) Suppose Bobby and Arnold could negotiate at zero transaction cost. What would be the outcome of the negotiations according to the Coase Theorem?

As the equilibrium is already KH-efficient, there is no need for negotiation. Hence, the result would remain the same.

(f) Suppose that Arnold and Bobby meet a wizard. The wizard casts a spell on all consumers, causing them to see all cars as if they were the same size. That is, consumers can no longer differentiate between large cars and small cars. However, consumers still value large cars differently from small cars. Explain why this may lead to a new equilibrium, in which some types of cars (either large, or small, or both) are not sold.

To get full points, the answer needs to explain how the adverse selection problem applies here (e.g. asymmetric information implies consumers will only be willing to pay their expected utility, and that WTP may be lower than the production cost of the higher-quality product, causing it to drop out of the market).

Q2. Bill and George each own an ice cream stand. Each of them are trying to decide whether or not to offer Caramel-flavored ice cream. Suppose that their current revenues are 100\$ each, and that if both of them decide offer this flavor, then their revenues will not change. However, if one of them offers the new flavor and the other one does not, then the one who offers the flavor will increase his revenue to 160\$ and the other one will get a lower revenue of only 60\$. Assume that the cost of offering the Caramel flavor is 10\$.

- (a) What is the Nash Equilibrium of this game if they both decide simultaneously?

Note: The cost of 10\$ for offering Caramel is part of the payoff and should be accounted for. Both players have a dominant strategy of offering Caramel: they earn $90 > 60$ if the other one also offers caramel, and $150 > 100$ if the other one does not. Hence, the equilibrium is (Caramel, Caramel). One can also draw a payoff table [but this was not required]:

	Caramel	No
Caramel	90,90	150,60
No	60,150	100,100

- (b) Is the Nash Equilibrium Pareto efficient? Is it Kaldor-Hicks efficient?

The NE is neither Pareto efficient nor KH efficient: The KH efficient outcome is that one offers caramel and the other does not (yielding a total of $150+60=210$). The NE, of (90,90), is the only that is not Pareto efficient: one could switch to the (100,100) where both don't offer Caramel and both would be better-off.

- (c) Compare this game to a Prisoner's dilemma. What is different and what is similar?

The game is quite similar to a prisoner's dilemma: both players play their dominant strategies and end up in a Pareto and KH inefficient outcome. The only difference is that here it is efficient that one offers Caramel and the other doesn't (whereas in the PD, if one confesses and the other does not it is KH inefficient) [students who noted the fact that payoffs were negative received full points – this is not exactly the issue, but close enough].

- (d) Suppose that Bill would decide first, and only then George will decide (after observing what Bill decides). What would be the Nash Equilibrium then?

Because the equilibrium is in dominant strategies, it makes no difference whether the game is sequential or static – the equilibrium remains the same (Caramel, Caramel).

- (e) Suppose that Bill and George can negotiate at zero cost. What would be the outcome of the game then? Would there be any money transfers between the parties?

According to the Coase Theorem, when TC are low (which is the case when they are zero), the parties will reach an ex-post efficient outcome through negotiations. Here, this means that the parties will coordinate on one producing Caramel and the other not – i.e on the KH efficient outcome. The person giving up their production of Caramel would only agree to

do this if he gets at least as much as in the original NE, i.e. at least 90\$. This means that whoever produces Caramel would need to pay at least 30\$ to the other one (resulting in $60+30=90$ \$). Note: the surplus from switching from the NE to KH-efficient outcome also happens to be 30 ($210-180=30$) but this is not the same 30\$ from above. The parties will split this surplus of 30, but not necessarily equally (the division depends on bargaining power).

Q3. Mary produces communication devices of two kinds: “A” and “B”. A is worth 100 to consumers, whereas B is only worth 50. However, consumers cannot differentiate between A and B, because they both look exactly the same. Mary is the only producer in the market, and she expects consumers to want 50 products for a price of 20\$, but decides to produce only 15 products and sell them for 40\$. Furthermore, the devices make a lot of noise when they are produced. One day, Mary asks a celebrity to recommend her devices, and suddenly everyone buys her devices.

Explain how each of the market failures covered in the course may manifest in market in which Mary operates in.

Market power: Mary is a monopolist (=possibly higher price inducing inefficiently low quantity).

Asymmetric information (adverse selection): consumers can't differentiate between the devices, causing consumers not to buy the good products.

Externalities: devices make a lot of noise (=discomfort to third parties).

Behavioral market failure (herding): the celebrity recommendation causes people to buy even if it is inefficient to do so.